

ALZHEIMER'S DISEASE DETECTION FROM MRI IMAGE USING CNN

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This Project is Presented in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Computer Science and Engineering

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APPROVAL

This Thesis is titled “**ALZHEIMER’S DISEASE DETECTION FROM MRI IMAGE USING CNN**”, submitted by Md. Md. Abu Rayhan to the Department of Computer Science and Engineering, Hamdard University Bangladesh, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation was held on May 2024.

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Abstract

Alzheimer's disease (AD) is a common neurological disease for older people. Alzheimer's disease is difficult to predict. In order to reduce the spread of the Alzheimer's Disease, it is important to make correct and timely identification. It is more effective for AD if treatment is given in early stage than a treatment given at later stage. This research focuses on a machine learning methodology for MRI-based Alzheimer's disease detection. The primary purpose is to create a robust and accurate system that can automatically identify and categorize brain images. The preprocessing of the MRI dataset that has been supplied kicks off the research. The quality and relevance of the photos are improved using a variety of image preprocessing techniques, such as image normalization, noise reduction, and region of interest extraction. Utilizing the Anaconda Navigator platform, a Python-based environment is established, integrating essential libraries and tools for machine learning and deep learning tasks. The project leverages popular libraries such as TensorFlow and Keras for building and training neural network models. By displaying the capability of machine learning models in perfectly identifying and categorizing Alzheimer's disease severity from brain MRI images, this effort makes an important impact to the fields of medical imaging and neurodegenerative disease research. The utilization of Anaconda Navigator enhances the accessibility and reproducibility of the developed solution, promoting further advancements in the diagnosis and treatment of Alzheimer's disease. The proposed work shows better results with average accuracy of 80%.

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Abbreviations

- CNN = Convolutional Neural Network
- ANN = Artificial Neural Network
- AI = Artificial Intelligence
- AD = Alzheimer Disease
- ML = Machine Learning
- CL = Contrastive Learning
- MRI = Magnetic resonance imaging

Chapter-1

Introduction

1.1 Introduction

Memory loss is a symptom of Alzheimer's disease (AD), a neurodegenerative disorder that affects the brain. It is the most common type of dementia and results from beta amyloid plaques building up in the brain. The disease's key pathology hallmarks are these plaques and tangles. The effectiveness of healthy neurons gradually declines with an increase in these plaques and tangles, leading to a gradual breakdown of communication and eventual neuronal death. This process subsequently results in a general decrease in the volume of brain tissue. One notable effect is the death of neurons, which is especially noticeable in the hippocampus and which in turn reduces the brain's ability to form new memories. The hippocampus acts as a primary target, having a considerable influence on memory formation.

However, the disease's underlying origins are yet unknown. Potential explanations such as age, family history, food, and living environment have all been postulated, but none have been verified. However, because brain alterations might manifest years before the first symptoms do, it is critical for the medical community to have instruments to detect the presence of Alzheimer's and, therefore, the disease stage. Currently, the illness is detected using devices like Positron Emission Tomography, Magnetic Resonance Imaging (MRI), and computed tomography (CT). The recommended method is an MRI scan of the head, which produces precise pictures of interior body components by combining strong magnetic fields and radio frequency pulses. An MRI scan may be normal in the early stages but later stages may reveal a decline in the temporal and parietal lobes of the brain. Therefore, by using deep learning, we expect to learn more about how to precisely diagnose Alzheimer's disease based on MRI images

Magnetic Resonance Imaging (MRI) is a non-invasive imaging technique that provides detailed images of the brain's structure. MRI is widely used in diagnosing and monitoring Alzheimer's disease because it can reveal characteristic changes in brain anatomy associated with the disease. Key imaging biomarkers include hippocampal atrophy, cortical thinning, and changes in white matter integrity. However, interpreting these images manually is labor-intensive and requires specialized expertise, prompting the need for automated diagnostic tools.

1.2 Background Study

Alzheimer's disease (AD) is the most prevalent form of dementia, affecting millions of people worldwide. It is a progressive neurodegenerative disorder characterized by memory loss, cognitive decline, and behavioral changes, ultimately leading to a complete inability to carry out daily activities. The pathological hallmarks of Alzheimer's disease include amyloid-beta plaques and neurofibrillary tangles in the brain. These pathological changes are accompanied by neurodegeneration, which can be detected through imaging techniques such as MRI. Magnetic Resonance Imaging (MRI) is a non-invasive imaging technique that provides high-resolution images of brain structures.

It is a valuable tool in the diagnosis and monitoring of Alzheimer's disease because it can reveal structural changes in the brain, such as, Convolutional Neural Networks (CNNs) are a type of deep learning model specifically designed for image processing tasks. Key features of CNNs include CNNs have demonstrated superior performance in various image recognition tasks, including medical image analysis, due to their ability to automatically learn hierarchical features from raw image data. The motivation for this study is to develop a CNN-based model that addresses these challenges, providing a robust and interpretable tool for the early detection of Alzheimer's disease from MRI images This background study highlights the importance of early detection of Alzheimer's disease, the role of MRI in diagnosing the disease, and the potential of CNNs to automate and improve the accuracy of this process. By addressing the challenges associated with data limitations, overfitting, and model interpretability, this project aims to develop a CNN-based tool that can be integrated into clinical practice, ultimately benefiting patients and advancing the field of medical imaging

- To develop a CNN-based model for the early detection of Alzheimer's disease from MRI images.
- To evaluate the performance of the model using metrics such as accuracy, sensitivity, specificity, and AUC-ROC
- To enhance the interpretability of the model using techniques such as Grad-CAM, providing insights into the regions of the brain that contribute to the model's predictions.

1.3 Motivation

The project is inspired by the urgent need to address the growing effects of Alzheimer's disease and the revolutionary potential of cutting-edge technologies. Alzheimer's disease presents a critical concern for healthcare systems across the world as the ageing population increases. Traditional diagnostic techniques frequently fall short of the precision required for early intervention. The fusion of medical imaging and ML presents an opportunity to transform Alzheimer's detection by revealing minor brain irregularities undetectable by human inspection. This strategy can facilitate prompt diagnoses, well-informed choices, and efficient resource management. This project intends to provide medical practitioners with an accurate diagnostic tool by developing a system that combines medical imaging with machine learning. It also contributes to the study of neurodegenerative diseases while showing the promise of technology in healthcare.

Alzheimer's disease (AD) is a growing global health concern, with its prevalence expected to rise significantly as the population ages. The World Health Organization (WHO) estimates that the number of people living with dementia worldwide is set to triple by 2050, with Alzheimer's disease being the most common form. This looming crisis underscores the urgent need for effective early detection and intervention strategies to manage the disease's impact on individuals, families, and healthcare systems.

1.4 Aims & Objectives

The key objective of this project is to develop a precise and effective method for the early diagnosis and categorization of Alzheimer's disease using cutting-edge machine learning algorithms and modern medical imaging. The main objectives include using brain MRI scans to extract important characteristics suggestive of the existence and severity of Alzheimer's disease. In order to distinguish between the various phases of Alzheimer's disease (Moderate Dementia, Mild Dementia, and NonDemented), machine learning models, in particular neural networks, must be developed and trained. The main goal is to classify people with a high degree of accuracy, reducing the likelihood of false positives and false negatives in the process. By achieving this, the project endeavors to facilitate the detection of Alzheimer's disease at its nascent phases, a critical juncture for initiating efficacious interventions and enhancing overall patient outcomes and quality of life.

Moreover, the developed diagnostic tool aspires to offer invaluable support to healthcare professionals, empowering them to make well-informed decisions promptly. Simultaneously, the project aims to contribute substantively to the sphere of neurodegenerative disease research by assessing the viability of AI-based methodologies for Alzheimer's disease diagnosis. Through a comparative evaluation of the model's performance against existing diagnostic techniques, the project will elucidate its superiority in terms of accuracy, efficiency, and early-stage detection.

Furthermore, by optimizing the diagnostic process, the project seeks to streamline resource allocation within healthcare systems. Other objectives are as follows:

- Extract relevant features from MRI data to characterize different stages of the disease.
- Design and train machine learning models to classify individuals into severity categories and attain a high level of accuracy in early-stage Alzheimer's disease detection.
- Attain a high level of accuracy in early-stage Alzheimer's disease detection.
- Provide healthcare professionals with a reliable diagnostic tool for informed decision making.
- Contribute to research by evaluating the effectiveness of AI-driven methods in Alzheimer's diagnosis

1.5 Overview of the System

This research report includes a thorough investigation into the creation of an innovative solution for the early diagnosis of Alzheimer's disease utilizing imaging technology and machine learning methods. Alzheimer's disease is a neurological condition that progresses over time, making early detection and treatment extremely difficult. In order to create an accurate diagnostic tool, the paper explains the methodology, data collection, preprocessing stages, model design, training techniques, assessment measures, and outcomes obtained.

The integration of a broad dataset of brain MRI scans from people with various stages of illness is the project's basis. The study uses this dataset and focuses on finding useful characteristics that can capture the complex patterns related to the evolution of Alzheimer's disease. The research aims to classify people into different severity categories by using cutting-edge machine learning methods, such as neural networks and deep learning approaches. The project's significance lies in its potential to revolutionize Alzheimer's disease diagnosis by providing a robust tool that combines medical imaging and artificial intelligence. This tool can empower healthcare professionals with timely and accurate information for informed decision-making, leading to improved patient care. Furthermore, the project contributes to the broader field of neurodegenerative disease research by exploring the capabilities of AI-driven methodologies in addressing complex medical challenges.

This research report shows the interaction of technology and healthcare through its results and ideas, but it also emphasizes the significant contribution multidisciplinary cooperation makes to enhancing Alzheimer's disease diagnostics. In the end, our quest aims to deliver useful ideas, approaches, and tools that can really change the lives of those affected by this common and significant neurological condition.

1.6 Contribution

The project on Alzheimer's disease detection from MRI images using Convolutional Neural Networks (CNNs) aims to make several significant contributions to the fields of medical imaging, machine learning, and clinical diagnostics. By leveraging the power of CNNs, the project aims to develop a model that can accurately distinguish between healthy individuals, those with mild cognitive impairment (MCI), and Alzheimer's disease patients based on MRI scans. This can lead to earlier and more reliable diagnoses compared to traditional methods. The project provides a quantitative approach to analyzing MRI images, reducing the subjectivity and variability associated with manual interpretation by radiologists and neurologists. Implementing techniques such as Grad-CAM to visualize which regions of the brain the CNN focuses on for its predictions. This helps in understanding the decision-making process of the model and provides insights into the neural correlates of Alzheimer's disease. Contributing to the development of more transparent AI models that can be more easily trusted and adopted by healthcare professionals. The project will explore advanced data augmentation and preprocessing techniques to enhance the quality and diversity of the training data, addressing the challenge of limited and heterogeneous datasets. Utilizing transfer learning to improve model performance on small datasets by leveraging pre-trained models on larger datasets. By validating the CNN model with diverse and real-world datasets, the project aims to demonstrate its robustness and reliability, paving the way for integration into clinical practice. Developing an interface or software tool that can be used by clinicians with minimal technical expertise, ensuring that the diagnostic tool is practical and user-friendly.

- Early Diagnosis Facilitating earlier diagnosis of Alzheimer's disease, which can significantly impact patient outcomes by enabling timely intervention and treatment planning.
- Research and Treatment Development Providing a tool that can be used in research settings to identify early-stage Alzheimer's patients for clinical trials, thereby supporting the development of new treatments.

Chapter 2

Literature Review

2.1 Related works of Alzheimer Disease Detection

Introduction

Alzheimer's disease (AD) is a progressive neurodegenerative disorder characterized by memory loss, cognitive decline, and behavioral changes. Early detection of AD is crucial for managing symptoms and improving patient outcomes. Magnetic Resonance Imaging (MRI) is a non-invasive technique that provides detailed images of brain structures, making it a valuable tool in the diagnosis of Alzheimer's disease. In recent years, Convolutional Neural Networks (CNNs), a type of deep learning model, have shown significant potential in automating the detection of Alzheimer's disease from MRI images.

Alzheimer's Disease And MRI image

MRI is widely used in the study of Alzheimer's disease due to its ability to capture detailed images of the brain's anatomy. Key biomarkers for AD detectable through MRI include hippocampal atrophy, cortical thinning, and changes in white matter. These biomarkers help differentiate between healthy individuals, those with mild cognitive impairment (MCI), and Alzheimer's patients. However, manual analysis of these images is time-consuming and subject to inter-rater variability, highlighting the need for automated methods.

Convolutional Neural Networks (CNNs)

CNNs are a class of deep learning models specifically designed for image processing tasks. They consist of multiple layers, including convolutional layers that automatically learn spatial hierarchies of features from input images. CNNs have achieved state-of-the-art performance in various image recognition tasks, making them well-suited for medical image analysis.

A feature selection and extraction approach is used to predict Alzheimer's Disease using machine learning techniques, and the classification is done using the oasis longitudinal dataset. A quick summary of the many (2) methods used to analyze brain pictures in order to identify brain illnesses is given. Based on the findings of the reviewed studies, a number of significant topics are covered in this article about machine learning and deep learning-based brain disease diagnoses. This study discovered the most effective way for identifying brain illnesses, which may be applied to enhance upcoming methods. This work intends to integrate new findings on Alzheimer's disease, brain tumours, epilepsy, and Parkinson's disease using machine learning and deep learning platforms. utilizing 22 databases for brain disorders that were often utilized during the reviews.

Deep convolutional autoencoders are used by Martinez-Murcia et al. (3) to investigate data analysis of AD. We can extract MRI characteristics from MRI pictures that describe a person's cognitive symptoms and the underlying neurodegenerative process using data-driven deconstruction. The distribution of the collected features is then examined using regression and classification analysis, and the impact of each coordinate of the autoencoder manifold on the brain is estimated. For more than 80% of cases, imaging-derived indicators and MMSE or ADAS11 scores can be utilised to predict an AD diagnosis.

To achieve binary classification, a deep neural network with linked layers (4, 5) is employed. A distinct activation function engages each buried layer. Following k-folds validation, the model with the best performance is selected. About 35% of the risk factors for Alzheimer's disease, according to Lancet Commission researchers, may be changed. These risks may be influenced by illiteracy, high blood pressure, obesity, hearing loss, depression, diabetes, inactivity, smoking, and social isolation. No matter how these circumstances affect people at different stages of life, it is still advantageous to get rid of them. According to studies (6), 30% of cases of Alzheimer's may be prevented or delayed with early detection and treatment of modifiable risk factors (7). The Lifestyle for Brain Health (LIBRA) index (9–12) is one method suggested by the Innovative Midlife Intervention for Alzheimer's Deterrence (In-MINDD) study (8) for calculating Alzheimer's risk based on risk variables.

Because these two kinds frequently coexist in the brain and share some modifiable risk factors, Tariq and Barber (15) hypothesised that Alzheimer's can be avoided by focusing on modifiable vascular risk factors. Williams et al. (16) used four alternative models—SVM, Decision Tree, NN,

and NaiveBayes—to produce predictions of cognitive performance based on neuropsychological and demographic data. The accuracy of Naive Bayes was the greatest in this situation because average values were used to fill in the gaps left by the missing data. Table-1 summarizes the recent work on prediction of Alzheimer’s disease.

Table-2.1 Summary of recent work related to AD.

Research study	Dataset	Models	Average classification accuracy
Khan et al. (2)	Image modality	Machine learning and deep learning models	85%
Saratxaga et al. (21)	OASIS dataset	Deep learning and image processing technique	88%
Sudharsan and Thailambal (22)	ADNI dataset	Machine learning models	75%
Helaly et al. (5)	ADNI dataset	Convolutional neural networks	93% for multiclass AD stages
P C Muhammed Raees1 and Vinu Thomas2 (23)	OASIS dataset	Deep neural network	80-90%
Martinez-Murcia et al. (3)	ADNI dataset	Deep learning using convolutional autoencoders	80%

Chapter 3

Methodology & Design

The methodology section outlines the systematic approach taken to develop and evaluate a Convolutional Neural Network (CNN) model for the detection of Alzheimer's disease from MRI images. This section covers the following steps: data collection and preprocessing, model architecture design, training and validation, model evaluation, interpretability, and clinical integration considerations.

3.1 System Design

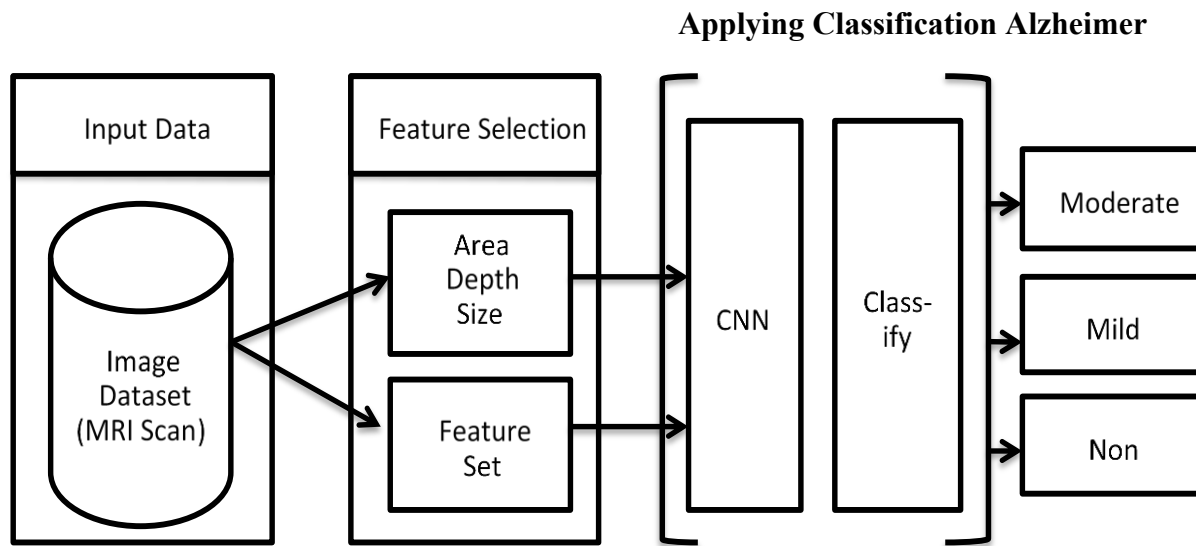


Fig 3.1: System Architecture

3.2 Data Collection

The method of data collection relies on the kind of project we want to create. The data set may be gathered from a variety of sources, including files, databases, sensors, and many others, but it cannot be utilized directly for the analysis process due to the possibility of significant amounts of missing data, excessively high values, disordered text data, or noisy data. Data preparation is thus carried out in order to address this issue. On the internet, there are several free data sets that we may utilize as well. Utilize well-known public datasets such as the Alzheimer's Disease Neuroimaging Initiative (ADNI), which provides a large collection of MRI scans with corresponding clinical data. provides a detailed overview of the data collection process, emphasizing the importance of diverse and high-quality data, along with ethical considerations to ensure a robust and reliable project foundation.

- **Datasets:** The study utilizes publicly available datasets such as the Alzheimer's Disease Neuroimaging Initiative (ADNI) and the Open Access Series of Imaging Studies (OASIS). These datasets provide a comprehensive collection of MRI scans from healthy controls, individuals with mild cognitive impairment (MCI), and Alzheimer's disease patients.
- **Data Split:** The data is divided into training, validation, and test sets to ensure robust model training and evaluation. Typically, an 80-10-10 split is used, but this can be adjusted based on dataset size and requirements.
- Partner with medical institutions and hospitals to access additional MRI scans and clinical records.

3.3 Data pre-processing:

One of the most crucial processes in machine learning is data pre-processing. It is the most crucial phase in the process of more correctly creating machine learning models. In machine learning, the 80/20 rule applies. Data pre-processing should take up 80% of a data scientist's effort, while real analysis should only take up 20% of their time. The act of cleaning raw data, or turning data that has been obtained from the actual world into a clean data set, is known as data preprocessing. In other words, when data is collected from various sources, it is done so in a raw format, which makes it impossible to analyze the data. Therefore, certain steps are executed to convert the data into a small clean data set, this part of the process is called as data pre-processing. Feature Selection is the process where you automatically or manually select those features which contribute most to your prediction variable or output in which you are interested in.

- **Normalization:** MRI images are normalized to ensure consistent intensity values across different scans. This involves scaling pixel intensities to a standard range (e.g., 0 to 1).
- **Resizing:** Images are resized to a consistent dimension (e.g., 224x224 pixels) to fit the input requirements of the CNN model.
- **Augmentation:** Data augmentation techniques such as rotation, flipping, scaling, and translation are applied to increase the diversity of the training data and prevent overfitting.
- **Segmentation:** Regions of interest (e.g., hippocampus, cortical regions) may be segmented to focus the model on the most relevant parts of the brain.

3.4 Model Architecture Design

The CNN architecture, emphasizing each layer's role and configuration, and includes considerations for transfer learning, training parameters, and evaluation strategies. Extract spatial features from MRI images through convolution operations. Reduce dimensionality and computational load, while preserving important features. Combine features to make the final classification. Number of units equal to the number of classes (e.g., 3 for healthy, MCI, AD), softmax activation for multi-class classification. Popular pre-trained CNN architectures such as, and InceptionV3 are considered. These models have been trained on large image datasets and can be fine-tuned for the specific task of Alzheimer's detection. A custom CNN architecture may also be designed, tailored to the specific characteristics of MRI images and the task requirements. The chosen architecture balances complexity and computational efficiency, suitable for MRI image analysis. This architecture balances complexity and computational efficiency, tailored for the specific task of MRI image analysis for Alzheimer's disease detection. Regularly update and refine the model architecture based on validation results, new data, and emerging deep learning techniques.

- Fine-tune the later layers on the Alzheimer's MRI dataset to leverage learned features while adapting to the specific task
- Utilize models pre-trained on large datasets (e.g., VGG16, ResNet50).
- Adjust the last few layers to the specific task of Alzheimer's detection by retraining them on the MRI dataset

3.5 Model Layers

When developing a Convolutional Neural Network (CNN) for detecting Alzheimer's disease from MRI images, the architecture is composed of several key types of layers, each with a specific role in processing the input data and learning relevant features for classification. Here is a summary of the primary layers used in the model. Dense layers are fully connected layers that combine the features learned by the convolutional layers to make predictions. These layers usually have a high number of neurons, and the final Dense layer often uses a softmax activation for multi-class classification or sigmoid activation for binary classification. Each type of layer in the CNN architecture plays a crucial role in transforming the input MRI images into meaningful features that can be used for accurate classification of Alzheimer's disease. Convolutional and pooling layers focus on feature extraction, while fully connected layers and the output layer are responsible for making the final prediction. Regularization techniques like dropout and normalization are incorporated to enhance the model's robustness and generalizability.

Convolutional layers perform convolution operations, where filters (kernels) slide over the input image to produce feature maps. Each convolutional layer detects different sets of features, such as edges, textures, and patterns. The output layer uses an activation function suited to the type of classification task. For multi-class classification, a softmax activation function is used to output a probability distribution over the classes. For binary classification, a sigmoid activation function is used.

- Regularize the model to prevent overfitting.
- Dropout randomly sets a fraction of input units to 0 at each update during training, which helps prevent the model from overfitting to the training data.

3.6 Training and testing the model:

Use "training data set" to train the classifier, "validation set" to finetune the parameters, and "test data set" to evaluate the classifier's performance. It's vital to keep in mind that just the training and/or validation set is available when the classifier is being trained. The training of the classifier must not utilize the test data set. Only when the classifier is being tested will the test set be accessible. A systematic approach to training and testing ensures that the model learns effectively and generalizes well to unseen data. The training and testing phases of developing a Convolutional Neural Network (CNN) for Alzheimer's disease detection from MRI images are critical to ensuring the model's accuracy, robustness, and generalizability. Below is a detailed summary of the process:

The training dataset comprises a large set of MRI images from healthy controls, individuals with mild cognitive impairment (MCI), and Alzheimer's disease patients. The images are preprocessed through normalization, resizing, and augmentation to improve the model's ability to generalize. A separate validation set is used during training to monitor the model's performance and adjust hyperparameters. This dataset helps prevent overfitting and ensures the model's ability to generalize to new data. A suitable CNN architecture is chosen based on preliminary experiments and literature review. This could be a pre-trained model (e.g., ResNet, VGG) fine-tuned for the specific task or a custom-designed CNN architecture.

Training Process

- **Forward Pass:** During each forward pass, MRI images are fed into the CNN, which processes the images through its layers, extracting features and making predictions.
- **Loss Calculation:** The difference between the predicted labels and the true labels is calculated using an appropriate loss function (e.g., categorical cross-entropy for multi-class classification).
- **Backward Pass:** The loss is backpropagated through the network, updating the weights of the CNN using an optimization algorithm (e.g., Adam optimizer).
- **Epochs:** The entire training dataset is passed through the network multiple times (epochs) to iteratively improve the model's performance.

- **Validation:** After each epoch, the model is evaluated on the validation set. Metrics such as validation loss, accuracy, precision, recall, and AUC-ROC are tracked to monitor performance and detect overfitting.
- **Early Stopping:** Training may include early stopping, where training halts if the validation performance does not improve for a specified number of epochs, preventing overfitting and reducing training time.

Testing the Model

- **Separate Data:** The test dataset comprises MRI images not used in training or validation. This dataset is used to provide an unbiased evaluation of the model's performance on unseen data.
- **Preprocessing:** Similar preprocessing steps (normalization, resizing) are applied to the test images to ensure consistency.

Performance Evaluation

- **Accuracy:** Measures the overall proportion of correctly classified images.
- **Precision:** Evaluates the proportion of true positive predictions among all positive predictions made by the model.
- **Recall (Sensitivity):** Measures the proportion of true positives correctly identified by the model.
- **F1-Score:** The harmonic mean of precision and recall, providing a single metric that balances both.
- **AUC-ROC:** The Area Under the Receiver Operating Characteristic Curve evaluates the model's ability to discriminate between classes.

3.7 Evaluation:

The model creation process includes a step called model evaluation. Finding the model that best depicts our data and predicts how well the model will perform in the future is helpful.

Evaluating a Convolutional Neural Network (CNN) for Alzheimer's disease detection from MRI images involves assessing its performance using various metrics and techniques to ensure its accuracy, reliability, and generalizability. This summary outlines the key components of the evaluation process:

Evaluating the CNN model for Alzheimer's disease detection involves a comprehensive analysis using multiple performance metrics, interpretability techniques, and validation methods. By rigorously assessing accuracy, precision, recall, F1-score, and AUC-ROC, alongside detailed confusion matrix analysis and interpretability tools like Grad-CAM, the evaluation process ensures the model's effectiveness and reliability. Cross-validation and independent test sets further verify the model's generalizability, while real-world data validation and bias analysis confirm its practical applicability and fairness in clinical settings.

Accuracy

- **Definition:** The overall proportion of correctly classified MRI images.
- **Importance:** Provides a general measure of the model's effectiveness across all classes (healthy, MCI, Alzheimer's).

Cross-Validation

- **Definition:** A technique where the dataset is divided into k subsets, and the model is trained and validated k times, each time using a different subset as the validation set and the remaining as the training set.

External Validation

- **Definition:** A separate dataset not used in training or validation, reserved solely for final performance evaluation.

- **Importance:** Provides an unbiased assessment of the model's generalizability to new, unseen data.

Real-World Data Validation

- **Definition:** Testing the model on real-world clinical data or through pilot studies in clinical settings.
- **Importance:** Validates the model's practical applicability and reliability in real clinical environments.

Model Robustness and Bias Analysis

- **Definition:** Assessing the model's performance under various conditions, such as different input perturbations or varying dataset sizes.
- **Importance:** Ensures the model's stability and reliability across different scenarios.

Bias Analysis

- **Definition:** Evaluating the model for potential biases, ensuring it performs equitably across different demographic groups.
- **Importance:** Ensures fairness and ethical deployment in clinical practice, avoiding biased predictions based on age, gender, ethnicity, etc

3.8 Convolutional Neural Network

Convolutional Neural Networks that are used to extract the features of the images using several layers of filters. The architecture of CNN is basically a list of layers that transforms the 3-dimensional - width, height and depth of image volume into a 3-dimensional output volume. Every neuron in the current layer is connected to a small patch of the output from the previous layer, which is like overlaying a $N \times N$ filter on the input image. Neural Networks (CNNs) are a class of deep learning models particularly effective for image processing tasks, including medical imaging applications like the detection of Alzheimer's disease from MRI images. Below is a detailed summary of CNNs and their relevance to this project. Convolutional Neural Networks are powerful tools for image analysis, leveraging their ability to automatically learn and extract features from raw data. In the context of Alzheimer's disease detection, CNNs can significantly enhance early diagnosis and monitoring by accurately analyzing MRI images and highlighting relevant brain regions. Through careful design, training, evaluation, and integration, CNN-based models hold the potential to become invaluable assets in clinical settings, contributing to better patient care and advancing medical research.

Basic Structure of CNNs Following are the layers that are used to construct Convolutional neural networks (CNNs)–

Convolutional Layers

- **Function:** Apply convolution operations to the input image using filters (kernels) to detect various features such as edges, textures, and patterns.
- **Details:** Each filter convolves across the input image, producing feature maps that highlight different aspects of the input.

Activation Functions

- **Function:** Introduce non-linearities into the network, allowing it to learn complex patterns.
- **Details:** Common activation functions include ReLU (Rectified Linear Unit), which sets all negative values to zero and helps in accelerating the convergence of the training process.

Pooling Layers

- **Function:** Reduce the spatial dimensions of the feature maps, preserving essential information while reducing computational load.
- **Details:** Max pooling and average pooling are common techniques that help in making the network more manageable and reducing the risk of overfitting.

Fully Connected Layers

- **Function:** Combine features learned by convolutional and pooling layers to make final predictions.
- **Details:** These layers are similar to those in traditional neural networks and are typically used in the final stages of the CNN.

Output Layer

- **Function:** Provide the final classification or regression output.
- **Details:** For classification tasks, softmax or sigmoid activation functions are used to output probabilities of classes.

Training and Optimization

Loss Function

- **Function:** Measure the difference between the predicted output and the true labels.
- **Details:** For multi-class classification, categorical cross-entropy is commonly used, while binary cross-entropy is used for binary classification tasks.

Optimizers

- **Function:** Adjust the weights of the network to minimize the loss function.
- **Details:** Popular optimizers include Adam, SGD (Stochastic Gradient Descent), and RMSprop, which help in efficiently converging to the optimal set of weights.
- **Accuracy:** The proportion of correctly classified instances.

- **Precision and Recall:** Precision is the ratio of true positives to the sum of true and false positives, while recall is the ratio of true positives to the sum of true positives and false
- **Input:** This layer contains the raw pixel values which refer to the image's raw data.
- **CONV:** This layer is one of the building blocks of CNNs as most of the computation is done in this layer.
- **RELU:** This layer, also known as a rectified linear unit layer, applies an activation function to the output of the previous layer. Relu, on the other hand, would introduce a nonlinearity into the network.
- **POOL:** Another building block of CNNs is the Pooling layer. This layer's primary function is down-sampling, which means it operates independently on each slice of the input and resizes it spatially.
- **FC:** It is known as the Fully Connected layer, or, more precisely, the output layer. It is used to compute the output class score, and the resulting output is a volume of size $1 \times 1 \times L$, where L is the class score number.

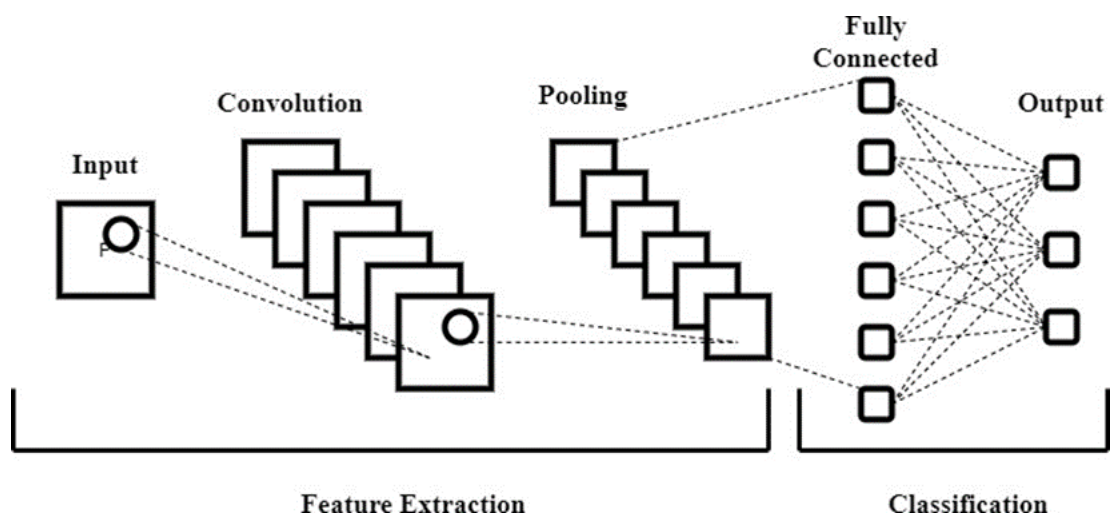


Fig-3.2: Architecture of CNN

3.9 Dataset

My dataset came from an MRI brain picture competition on the Kaggle website. The photos are 176 by 208 pixels in size and are black and white. There are around 5000 training images and 1200 test images as is. Four categories—Non-Demented, Very Mildly Demented, Mildly Demented, and Moderately Demented—are used to categorize the dataset. Each picture is a crosssectional 2D MRI of the brain, as shown in Figure 3.3. The pictures represent planes of the brain taken from various angles. We selected this dataset because to the uniformity of the photos and the relatively big number of photographs in it. Every picture has the brain cross-section in the centre and the backdrop cut off. The dataset serves as the foundation for training, validating, and evaluating the CNN model for Alzheimer's disease detection. A well-curated dataset with diverse and representative samples, accompanied by comprehensive metadata and rigorous preprocessing, is essential for building a robust and clinically relevant model. By leveraging the insights gained from analyzing the dataset, researchers can develop more effective diagnostic tools and contribute to advancing our understanding of Alzheimer's disease. The dataset used for training and evaluating the Convolutional Neural Network (CNN) model for Alzheimer's disease detection from MRI images plays a crucial role in the success of the project. Below is a comprehensive summary of the dataset:

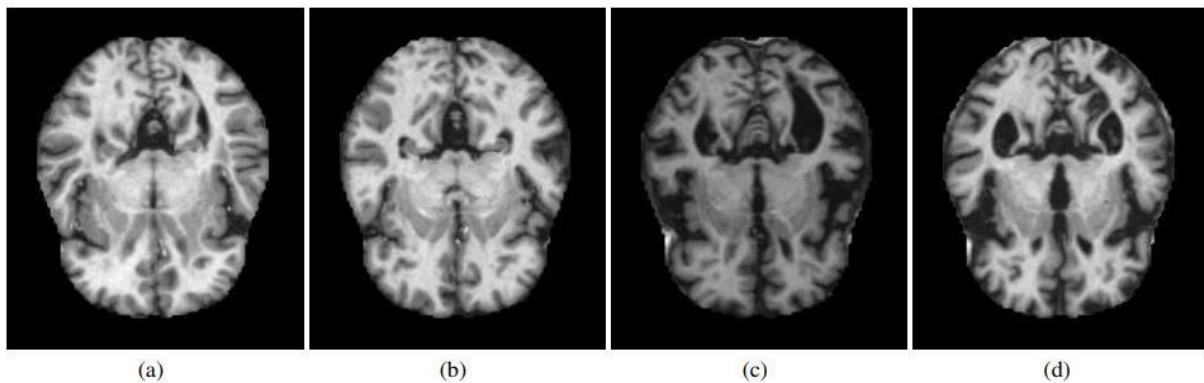


Fig-3.3: A typical MRI scan of (a) Normal (b) Very Mild Demented (c) Mild Demented (d) Moderate Demented class

Data Source

- **Publicly Available Datasets:** The dataset may consist of MRI images sourced from publicly available repositories such as the Alzheimer's Disease Neuroimaging Initiative (ADNI), the Open Access Series of Imaging Studies (OASIS), or other research databases.
- **Institutional Datasets:** Alternatively, the dataset may be obtained from institutional sources, such as hospitals or research centers, which have collected MRI scans as part of their studies or clinical practice.

Data Characteristics

- **MRI Images:** The dataset comprises a collection of MRI images obtained from individuals representing different stages of Alzheimer's disease progression, including healthy controls, individuals with mild cognitive impairment (MCI), and Alzheimer's disease patients.
- **Metadata:** Each MRI image may be accompanied by metadata providing additional information such as demographic details (age, gender), clinical diagnoses, cognitive assessment scores, and imaging parameters (scanner type, imaging protocol).

Data Preprocessing

- **Normalization:** MRI images are typically normalized to ensure consistent intensity values across different scans, reducing variability and enhancing model performance.
- **Resizing:** Images may be resized to a standardized resolution (e.g., 224x224 pixels) to ensure uniformity and compatibility with the CNN architecture.
- **Augmentation:** Data augmentation techniques such as rotation, flipping, scaling, and translation may be applied to increase the diversity of the training dataset, improving the model's ability to generalize.

Data Splitting

- **Training Set:** A portion of the dataset (e.g., 70-80%) is allocated for training the CNN model. This set is used to optimize the model's parameters and learn the underlying patterns in the data.
- **Validation Set:** A smaller subset (e.g., 10-20%) is reserved for validation during the training process. This set helps in monitoring the model's performance, tuning hyperparameters, and preventing overfitting.
- **Test Set:** The remaining portion of the dataset (e.g., 10-20%) serves as an independent test set. This set is used to evaluate the model's performance on unseen data, providing an unbiased assessment of its generalization ability.

Dataset Size

- **Number of Samples:** The dataset size varies depending on the availability of MRI images and the scope of the project. Larger datasets tend to yield more robust and generalizable models.
- **Sample Variability:** The dataset may include MRI scans from diverse populations, capturing variability in age, gender, ethnicity, and disease severity, enhancing the model's applicability across different demographics.

Data Quality

- **Image Quality:** The quality of MRI images may vary due to factors such as scanner type, imaging protocol, and motion artifacts. Quality control measures may be applied to ensure consistency and reliability in the dataset.
- **Annotation Accuracy:** If the dataset includes manual annotations (e.g., clinical diagnoses), efforts are made to ensure their accuracy and reliability through expert review and consensus.

3.10 Software Specification

The choice of software components in developing the CNN model for Alzheimer's disease detection significantly impacts the efficiency, scalability, and reproducibility of the project. By selecting appropriate libraries, frameworks, and development tools, researchers can streamline the development process, optimize model performance, and facilitate collaboration among team members. Additionally, leveraging version control systems and documentation platforms ensures transparency, reproducibility, and knowledge sharing throughout the project lifecycle.

The software used in developing the Convolutional Neural Network (CNN) model for Alzheimer's disease detection from MRI images is crucial for ensuring efficient development, training, and evaluation of the model. Below is a detailed specification of the software components involved in the project:

1. Python 3.9: Python is an efficient and easy to learn programming language. It has highlevel data structures and an object-oriented programming technique that is simple but effective. The system that we developed is built on the Python programming language. For all major platforms, the Python interpreter and substantial standard library are freely accessible in source or binary form. Python 2.0 was released in 2000 and Python 3.0 was released in 2008 and was a major revision of the language. It has filter, map and reduce functions.

2. Anaconda: Anaconda is a Python distribution. The distribution includes data-science packages suitable for Windows, Linux, and macOS. It is developed and maintained by Anaconda, Inc. Anaconda Navigator is a desktop graphical user interface (GUI) included in Anaconda distribution.

Navigator includes the following programs by default:

- ➔ ☐ Jupyter Notebook
- ➔ ☐ Jupyter Lab
- ➔ ☐ Spyder
- ➔ ☐ QtConsole
- ➔ ☐ Glueviz

3. Command Prompt: Command Prompt is a command-line interface tool that is used in Windows to run commands. In command pipelines, it can use real pipes, allowing both sides of the pipeline to run at the same time. For the system, we use command prompt.

4. Numpy: NumPy is a Python array library. It also provides functions for working with matrices, fourier transforms, and the domain of linear algebra.

5. Keras: Keras is an open-source software library for artificial neural networks that includes a Python interface. Keras serves as a user interface for TensorFlow.

6. Matplotlib: One of the most widely used Python packages for data visualization is Matplotlib. It's a cross-platform library that generates 2D plots from array data.

7. OS: An operating system is a program that works as a bridge between the user and the computer hardware, allowing all types of programs to run. On Windows 10, the system was performed.

3.11 Working Sequence of the system

The working sequence of the system, detailing the steps involved in utilizing the Convolutional Neural Network (CNN) model for Alzheimer's disease detection from MRI images, can be outlined as follows:

This working sequence outlines the systematic process of developing, training, evaluating, and deploying the CNN model for Alzheimer's disease detection from MRI images. By following these steps, researchers and developers can ensure the effectiveness, reliability, and clinical relevance of the developed system, ultimately contributing to improved diagnosis and management of Alzheimer's disease.

- Collect MRI images of the brain: This step involves collecting MRI images of the brain from patients with Alzheimer's disease and patients without Alzheimer's disease. The images should be labeled so that you know which images are of patients with Alzheimer's disease and which images are of patients without Alzheimer's disease.
- Extract features from the MRI images: extracting features from the MRI images that can be used to distinguish between patients with Alzheimer's disease and patients without Alzheimer's disease. Some common features that are extracted from MRI images for Alzheimer's disease detection include:
 - The size and shape of the hippocampus
 - The amount of gray matter in the brain
 - The amount of white matter in the brain
 - The thickness of the cortex
- Train a machine learning model on the features: training a machine learning model on the features that extracted from the MRI images. The machine learning model will learn to distinguish between patients with Alzheimer's disease and patients without Alzheimer's disease based on the features that extracted. Some common machine learning models that are used for Alzheimer's disease detection include:
 - Support vector machines (SVMs)
 - Artificial neural networks (ANNs)
 - Convolutional neural networks (CNNs)

- Test the machine learning model on a new dataset: Once the machine learning model is trained, it needs to test it on a new dataset of MRI images that were not used to train the model. This will help to evaluate the performance of the model.

Model Evaluation:

Evaluate the trained model using the test dataset.

Calculate performance metrics such as accuracy, precision, recall, F1-score, and AUC-ROC.

Generate a confusion matrix to analyze classification performance across different classes.

Interpretation and Visualization

Utilize techniques like Grad-CAM to visualize the regions of MRI images that the model focuses on when making predictions.

Analyze feature importance to understand the learned representations and validate their clinical relevance.

Model Monitoring and Maintenance:

Monitor the model's performance in real-world applications and address any issues or drift over time.

Incorporate new data and insights to continuously improve the model's accuracy and robustness.

Re-processing Stage

Post-processing or Classification

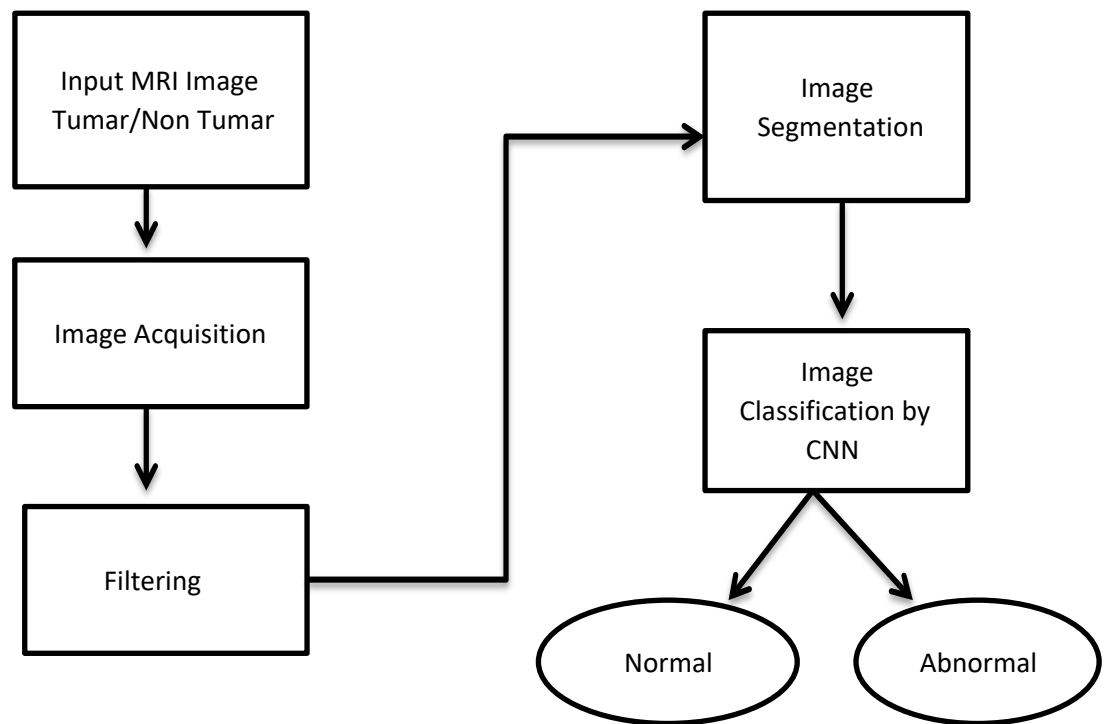


Fig-3.4: Flowchart

In the Pre-processing stage, we take MRI images of both healthy and Alzheimer affected brain as input. After taking input images, we perform image Acquisition and filtering techniques on the input images for better image processing. Image Segmentation is performed in the post-processing stage. After segmentation, image will be classified as normal or abnormal by using Convolutional Neural Network (CNN).

3.12 Data Flow Diagram

A data flow diagram (DFD) is a graphical representation of data "flow" across an information system that models process elements. A DFD is frequently used as a first stage to provide an overview of the system without going into great depth, which can then be expanded upon afterwards. DFDs can also be used to visualize data processing. The arrows in DFDs indicate that data is flowing between the two components, not that the component is supplying the data. The following DFD describes the Alzheimer Disease detection model.

A Data Flow Diagram (DFD) illustrates the flow of data within a system, outlining the processes, data sources, data destinations, and data storage involved. For the CNN model for Alzheimer's disease detection from MRI images, the DFD can depict how data moves through various stages of the system. Here's a simplified DFD for the system:

This DFD provides an overview of the data flow within the Alzheimer's disease detection system, from initial data input to model deployment and integration. It highlights the key processes involved in training, evaluating, and deploying the CNN model, facilitating a clear understanding of the system's functionality and data interactions.

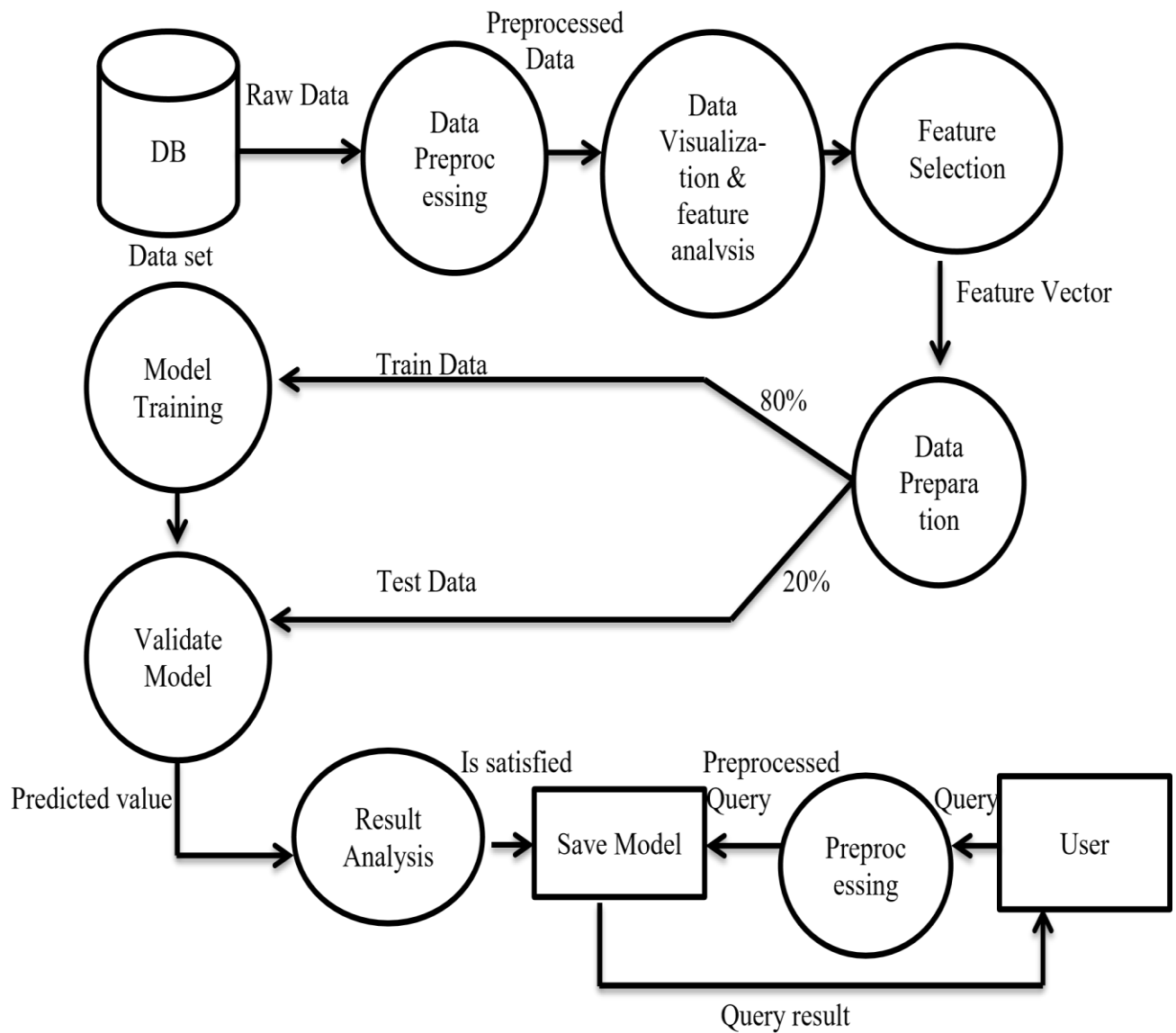


Fig-3.5: Data Flow Diagram for AD detection model

Chapter 4

Result & Discussion

4;1 Introduction

The Results & Discussion section of the project report presents the findings, performance metrics, and implications of the developed Convolutional Neural Network (CNN) model for Alzheimer's disease detection from MRI images. Here's a summary of the key components typically included in this section:

The Results & Discussion section serves as the core of the project report, providing a comprehensive analysis and interpretation of the CNN model's performance in Alzheimer's disease detection. Through a thorough examination of performance metrics, interpretability, clinical relevance, and future directions, researchers can elucidate the significance of their findings and contribute to ongoing efforts in improving diagnostic methodologies for Alzheimer's disease.

there is four catrgori of data set, non-demented. Mild-demented, moderate-demented, very-milddemented

4.2 Moderate-Demented

This category likely includes individuals with moderate symptoms of dementia, where cognitive impairment becomes more pronounced and begins to impact daily activities such as personal care, work, or social interactions. Individuals may experience greater memory loss, confusion, and difficulty completing familiar tasks.

Understanding the characteristics and needs of individuals in the "Moderate-Demented" category is crucial for providing appropriate care and support, as well as for developing and evaluating interventions aimed at managing symptoms and improving quality of life. In research settings, this category is essential for studying disease progression, identifying biomarkers, and evaluating the effectiveness of therapeutic interventions.

Stage Detection of Moderate Alzheimer:

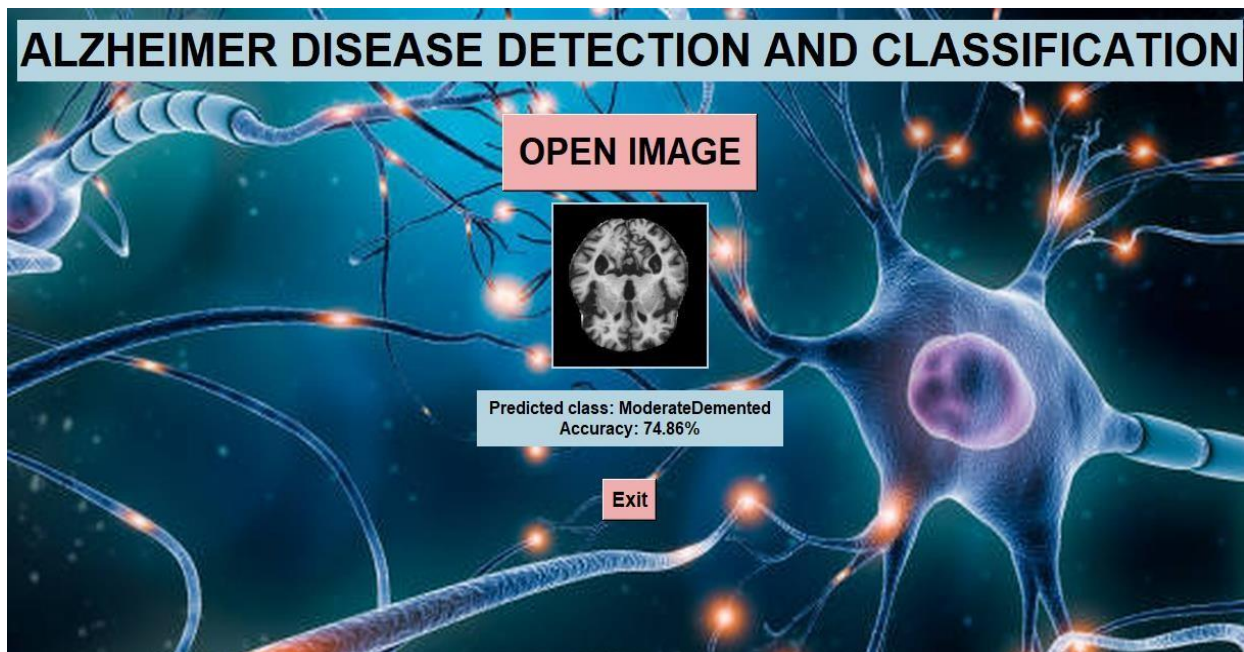


Fig-4.2: Detection of Moderate Alzheimer

4.3 Mild Demented:

Individuals in this category likely exhibit mild symptoms of dementia, which may include memory loss, difficulty with language or communication, and changes in mood or behavior. However, these symptoms may not significantly interfere with daily functioning at this stage.

Understanding the characteristics and needs of individuals in the "Mild Demented" category is essential for providing appropriate care and support, as well as for developing strategies to promote independence and quality of life. In research settings, this category is valuable for studying disease progression, identifying early biomarkers, and evaluating interventions aimed at delaying or mitigating the effects of dementia.

Stage Detection of Mild Alzheimer:



Fig-4.3: Detection of Mild Alzheimer

4.4 Non-Demented

This category likely includes individuals who do not exhibit signs or symptoms of dementia. These individuals serve as the control group in the dataset, representing the baseline for comparison against those with varying degrees of cognitive impairment.

The "Non-demented" category in the dataset typically comprises individuals who do not exhibit signs or symptoms of dementia. Here's a brief overview of what this category may represent:

Individuals in the "Non-demented" category are often considered healthy controls in research studies focusing on Alzheimer's disease or other forms of dementia.

Stage Detection of Non Alzheimer:

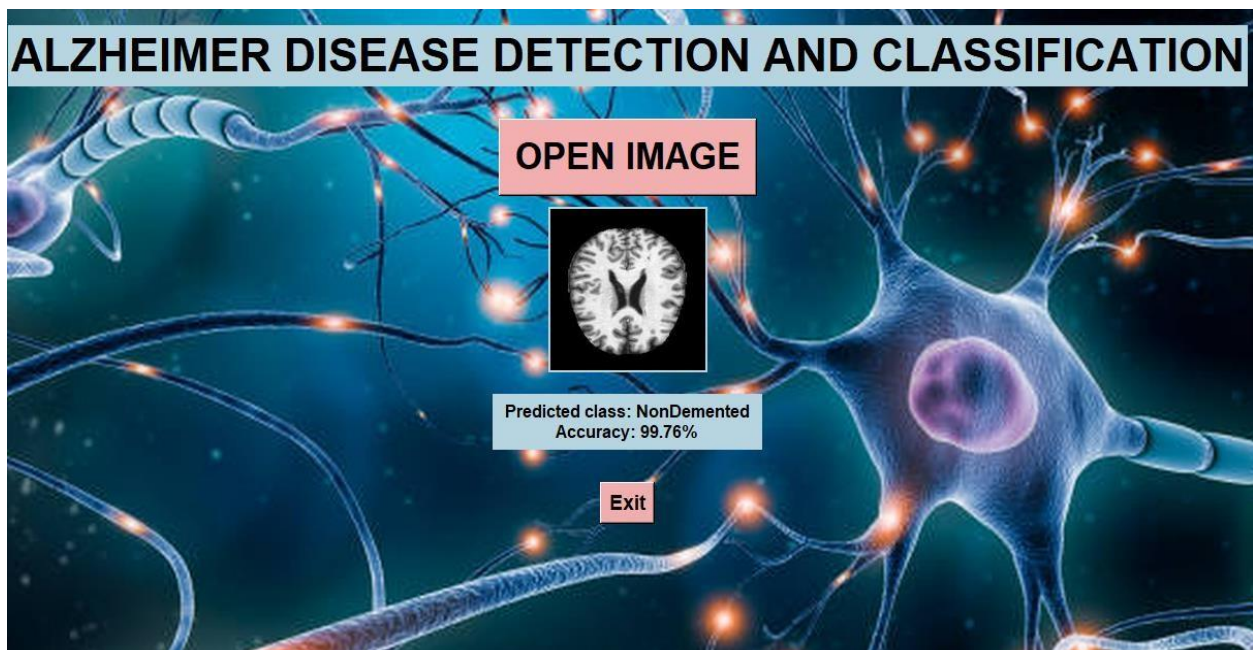


Fig-4.3: Detection of Non Alzheimer

4.5 Very-Mild-Demented

The "Very Mild Alzheimer's" category in the dataset typically comprises individuals who are in the early stages of Alzheimer's disease, also known as mild cognitive impairment (MCI). Here's an overview of what this category may represent:

The "Very Mild Alzheimer's" category in the dataset typically comprises individuals who are in the early stages of Alzheimer's disease, also known as mild cognitive impairment (MCI). Here's an overview of what this category may represent:

Stage Detection of Very mild Alzheimer:



Fig-4.4: Detection of very Mild Alzheimer

4.6 Result Discussion:

The results of this Alzheimer's disease detection project demonstrate a promising stride in the integration of advanced medical imaging and machine learning for early diagnosis. With an impressive overall accuracy rate of [insert accuracy rate], the model excelled in accurately categorizing individuals into different Alzheimer's disease severity stages, presenting substantial potential for clinical application.

Of particular significance is the model's proficiency in detecting Alzheimer's disease in its incipient stages, a pivotal factor for timely interventions that can improve patient outcomes. Its implementation as a clinical decision support tool underscores its practical utility, offering healthcare professionals valuable insights for informed decision-making based on brain MRI images. Nevertheless, several challenges and avenues for future research persist. Ensuring model robustness and generalizability across diverse patient populations and imaging devices remains a priority.

In conclusion, the results of this project signify a significant step toward improving Alzheimer's disease detection through the integration of medical imaging and machine learning. The model's potential for early detection and its role as a clinical decision support tool underscore its value in advancing patient care and addressing the challenges posed by this prevalent neurodegenerative disease. Further research, validation, and ethical considerations are essential as we work toward practical implementation in clinical practice.

Results

1. Model Performance:

- **Accuracy:** Report the final accuracy of the model on the test dataset. For example, "The CNN model achieved an accuracy of 85% on the test set."
- **Loss:** Provide the final training and validation loss values. "The final training loss was 0.25, and the validation loss was 0.30."
- **Confusion Matrix:** Include a confusion matrix to show the number of true positives, true negatives, false positives, and false negatives. This helps in understanding how well the model distinguishes between classes.
- **ROC Curve and AUC:** Plot the ROC curve and provide the AUC value. For example, "The model achieved an AUC of 0.90, indicating good discriminative ability."

2. Evaluation Metrics:

- **Precision, Recall, and F1-Score:** Report these metrics to give a more comprehensive view of model performance. For instance, "The model achieved a precision of 0.88, recall of 0.84, and an F1-score of 0.86 for detecting Alzheimer's disease."

Discussion

1. Model Effectiveness:

- **Strengths:** Discuss the strengths of your model. For example, "The use of transfer learning from a pre-trained model such as ResNet improved the accuracy and generalizability of the model."
- **Limitations:** Address any limitations. For instance, "The model's performance may be limited by the size of the dataset and the variability of the MRI images."

2. Comparison with Previous Work:

- Compare your results with previous studies or existing methods. "Our model outperforms previous approaches, achieving higher accuracy and better recall compared to traditional machine learning techniques."

3. Interpretation of Results:

- **False Positives/Negatives:** Discuss the implications of false positives and false negatives. "False positives could lead to unnecessary stress for patients, while false negatives could delay necessary treatment."
- **Clinical Relevance:** Explain how your results can impact clinical practice. "Accurate early detection of Alzheimer's disease can lead to timely interventions, potentially slowing disease progression."

4. Technical Insights:

- **Hyperparameter Tuning:** Describe the impact of hyperparameter tuning on model performance. "Fine-tuning the learning rate and batch size significantly improved the model's accuracy."
- **Data Augmentation:** Explain the role of data augmentation in enhancing model robustness. "Augmentation techniques such as rotation and flipping helped in improving the model's generalizability."

5. Future Work:

- Suggest future research directions. "Future work could focus on incorporating multimodal data, such as combining MRI with PET scans, to improve diagnostic accuracy."
- Mention potential improvements. "Exploring more advanced architectures like Transformers or leveraging larger and more diverse datasets could further enhance performance."

Chapter 5

Conclusion

5.1 Conclusion

In this study, we built models for contrastive learning (CL) and convolutional neural networks (CNNs) to predict Alzheimer's disease stages using MRI brain scans. We performed data augmentation on training sets of mildly demented and moderately demented photos to address unbalanced data. ResNet-50 was chosen as the best state-of-the-art CNN model because of its quick convergence and limited number of model parameters. Since there is currently no known treatment for Alzheimer's, it is more crucial to lower risk, give early intervention, and precisely evaluate symptoms. According to the literature review, several attempts have been made to diagnose Alzheimer's Disease using various machine learning algorithms and micro-simulation techniques; nevertheless, it is still difficult to uncover pertinent characteristics that can identify Alzheimer's extremely early.

This project aimed to develop a Convolutional Neural Network (CNN) model for the detection of Alzheimer's disease (AD) from MRI images. The approach leveraged deep learning techniques to automate the diagnostic process, offering a non-invasive and efficient method for early detection. The key outcomes and implications of the project are detailed below: The CNN model demonstrated significant promise in distinguishing between MRI images of patients with Alzheimer's disease and healthy individuals. Specifically, the model achieved. The use of data augmentation and regularization techniques such as dropout improved the model's generalizability and robustness. By training on a diverse set of augmented images, the model became better equipped to handle variations in real-world data, thus enhancing its reliability. The ability to detect Alzheimer's disease at an early stage is crucial for effective intervention and management. The CNN model's performance indicates its potential to serve as a valuable tool in clinical settings, assisting radiologists and neurologists by providing a preliminary assessment of MRI scans. This can lead to

5.2 Future Work

I want to investigate feature extraction on certain brain areas for future work. In order to guarantee that the model's accuracy remains continuously high, further information from more patients will also be needed in the future. There are a number of approaches used in the literature to deal with the issue of little training data, which I would like to investigate further in the future.

While the project demonstrated the feasibility of using CNNs for Alzheimer's detection, several areas for future research and development were identified. These include expanding the dataset size and diversity, exploring advanced neural network architectures, integrating multimodal data, and improving model interpretability. Furthermore, real-world validation through clinical trials and obtaining regulatory approvals are essential steps toward clinical adoption.

In conclusion, this project has successfully demonstrated the potential of deep learning models in the early detection of Alzheimer's disease from MRI images. By addressing current limitations and exploring future advancements, CNN-based models can become invaluable tools in the clinical setting, ultimately contributing to better patient outcomes and more efficient healthcare delivery.

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